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A bunch of text that I didn't write, except for the abstract (which I wrote)

testing out titles and stuff

Not Me

2025-08-26

Abstract

Magnetism is a major research program within condensed matter physics, and much research, both experimental and theoretical, is dedicated towards understanding magnetism. An outsized portion of this research is dedicated towards ferromagnetism and antiferromagnetism, yet none of these appease Lord Sobek as he demands. Thus, in this paper, we demonstrate a novel magnetic phase, pharaohmagnetism, brought about by using an Ankh on a slab of iron. We also develop new theoretical methods for analyzing pharaohmagnetic materials using Anubis–Ma'at statistics, predicting a secondary class of *anti*pharaohmagnetic materials. I will surely win the nūb-el prize.

The University of Life

1. Hey, I didn't write this

The following text is a collection of *fragments* taken from Lieb, Schultz, Mattis (2004/1961).[1] Please, go read the actual paper, it's great.

1.1. Formulation

The first model consists of N spin 1/2's (N even) arranged in a row and having only the nearest neighbor interactions. It is

$$H_\gamma = \sum_i \left[(1+\gamma) S_i^x S_{i+1}^x + (1-\gamma) S_i^y S_{i+1}^y\right],$$

where γ is a parameter characterizing the usual degree of anisotropy in the xy-plane. Because the Hamiltonian only involves the x- and y- components of the spin operators, we call this model the XY model.

To solve the XY model, we first introduce the raising and lowering operators

$$a_i^\dagger = S_i^x + \mathrm{i}\, S_i^y \text{ and } a_i = S_i^x - \mathrm{i}\, S_i^y$$

in terms of which the Pauli spin operators are

$$S_i^x = \left(a_i^\dagger + a_i\right)/2;\; S_i^y = \left(a_i^\dagger - a_i\right)/2\mathrm{i};\; S_i^z = a_i^\dagger a_i - \frac{1}{2}$$

and the Hamiltonian is

$$H_\gamma = \frac{1}{2} \sum_i \left[\left(a_i^\dagger a_{i+1} + \gamma\, a_i^\dagger a_{i+1}^\dagger\right) + \text{h.c.}\right].$$

The ground state Ψ_0 is the state with no elementary excitations:

$$\eta_k \Psi_0 = 0\,, \text{ all } k\,.$$

The ground-state energy, according to (A-12) is

$$E_0 = -\frac{1}{2} \sum_k \Lambda_k\,.$$

In the limit $N \to \infty$, the sum can be replaced by an integral giving

$$\begin{aligned} \frac{E_0}{N} &= -\frac{1}{4\pi} \int_{-\pi}^{+\pi} \left[1 - \left(1-\gamma^2\right) \sin^2 k\right]^{1/2} dk \\ &= -\frac{1}{\pi}\, \mathcal{E}\left(1-\gamma^2\right), \end{aligned}$$

[1] Two Soluble Models of an Antiferromagnetic Chain. In: Nachtergaele, B., Solovej, J.P., Yngvason, J. (eds) Condensed Matter Physics and Exactly Soluble Models. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-662-06390-3_35

where $\mathcal{E}(k^2)$ is one of the complete elliptic integrals (13).

1.2. Some more stuff

1.2.1. Spin ice cream.

Yum yum.